

试错：

```
device/amlogic/common/sepolicy/system_app.te:93:WARNING 'unrecognized character' at token '+' on line 73648:
allow system_app mnt_media_rw_file:file { read write open getattr create };
+#[SE1-zanrry-2020/08/06]Add apk call libprovision { */
[ 985 417/423] build out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.sorted.tmp
FAILED: out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.sorted.tmp
/bin/bash -c "(out/host/linux-x86/bin/checkfc -e out/target/product/ohm/obj/ETC/sepolicy_intermediates/sepolicy out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.tmp ) && (out/host/linux-x86/bin/fc_sort -i out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.sorted.tmp )"
out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.tmp: line 741 has invalid file type u:object_r:xulantian_block_device:s0
out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.tmp: line 741 has invalid file type u:object_r:xulantian_block_device:s0
Error: could not load context file from out/target/product/ohm/obj/ETC/file_contexts.bin_intermediates/file_contexts.device.tmp
21:25:18 ninja failed with: exit status 1

#### failed to build some targets (01:53 (mm:ss)) ####
```

diff --git a/sepolicy/file_contexts b/sepolicy/file_contexts
+/cas(/.*)? u:object_r:cas_block_device:s0

定位：在make custom_images 中 某些路径是空的情况，报错如图：

```
vendor/sdmc/sdmc-libs/libmulti_wifi/multi_wifi_load/Android.mk:3: warning: BUILD_BCM_TCP_SH is
[ 99% 225/226] Build image out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/xulantian.img
FAILED: out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/xulantian.img
/bin/bash -c "(rm -rf out/target/product/ohm/obj/PACKAGING/xulantian_intermediates ) && (rm -rf out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/cas ) && (mkdir -p out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/cas ; cp -Rf device/amlogic/ohm/xulantian_intermediates/cas ; mkdir -p ; cp -Rf /xulantian.txt ) && (echo \"# For all accepted properties, see BuildImage() in tools/releasetools/build_image.py\" > out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"mount_point=cas\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"partition_name=cas\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"fs_type=ext4 CUSTOM_IMAGE_DICT_FILE := device/amlogic/ohm/xulantian/xulantian.dict.txt\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"partition_size=16777216\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"ext_mkuserimg=mksuserimg mke2fs\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"selinux_fc=out/target/product/ohm/obj/ETC/file_contexts.bin\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (echo \"avb_avbtool-out/host/linux-x86/bin/avbtool\" >> out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt ) && (PATH=out/host/linux-x86/bin:/system/extras/ext4_utils/;$PATH out/host/linux-x86/bin/build_image out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/cas out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/image_info.txt out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/xulantian.img out/target/product/ohm/system )"
mkdir: Needs 1 argument (see "mkdir --help")
cp: Needs 2 arguments
21:46:44 ninja failed with: exit status 1

#### failed to build some targets (01:23 (mm:ss)) ####

xulantian@rd24:~/workspace/s905x4/s/s905x4_sdk0805_reference_userdebug/sdmc/s-amlogic-source-20220805-gtvs.xml$
```

定位到：报错位置在build/core/task/toools/build_custom_image.mk 中的118-121行

```
113 @echo "Build image $@"
114 $(hide) rm -rf $(PRIVATE_INTERMEDIATES) && mkdir -p $(PRIVATE_INTERMEDIATES)
115 $(hide) rm -rf $(PRIVATE_STAGING_DIR) && mkdir -p $(PRIVATE_STAGING_DIR)
116 # Copy all the files.
117 $(hide) $(foreach p,$(PRIVATE_COPY_PAIRS),\
118     $(eval pair := $(subst :,$(space),$(p)))\
119     mkdir -p $(dir $(word 2,$(pair)))\
120     cp -Rf $(word 1,$(pair)) $(word 2,$(pair));)
121 $(if $(PRIVATE_PICKUP_FILES),$(hide) cp -Rf $(PRIVATE_PICKUP_FILES) $(PRIVATE_STAGING_DIR))
122 # Generate the dict.
123 $(hide) echo "# For all accepted properties, see BuildImage() in tools/releasetools/build_image.py" > $(PRIVATE_INTERMEDIATES)/image_info.txt
124 $(hide) echo "mount_point=$(PRIVATE_MOUNT_POINT)" >> $(PRIVATE_INTERMEDIATES)/image_info.txt
125 $(hide) echo "partition_name=$(PRIVATE_MOUNT_POINT)" >> $(PRIVATE_INTERMEDIATES)/image_info.txt
126 $(hide) echo "fs_type=$(PRIVATE_FILE_SYSTEM_TYPE)" >> $(PRIVATE_INTERMEDIATES)/image_info.txt
127 $(hide) echo "partition_size=$(PRIVATE_PARTITION_SIZE)" >> $(PRIVATE_INTERMEDIATES)/image_info.txt
128 $(hide) echo "ext_mkuserimg=$(notdir $(MKEXTUSERIMG))" >> $(PRIVATE_INTERMEDIATES)/image_info.txt
129 $(if $(PRIVATE_SELINUX),$(hide) echo "selinux_fc=$(SELINUX_FC)" >> $(PRIVATE_INTERMEDIATES)/image_info.txt)
130 $(if $(PRIVATE_SUPPORT_VERITY),\
```

```
124     echo "mkdir_path=$(dir $(word 2,$(pair)))"
125     echo "cp_path=$(word 1,$(pair)) $(word 2,$(pair));"
126     $(hide) $(foreach p,$(PRIVATE_COPY_PAIRS),\
127         $(eval pair := $(subst :,$(space),$(p)))\
128         mkdir -p $(dir $(word 2,$(pair)))\
129         cp -Rf $(word 1,$(pair)) $(word 2,$(pair));)
130     echo "p=$(p)"
131     echo "PRIVATE_COPY_PAIRS= $(PRIVATE_COPY_PAIRS)"
```

```

echo ${PRIVATE_COPY_PAIRS})

$(hide) $(foreach p,${PRIVATE_COPY_PAIRS},\
    $(eval pair := $(subst :, $(space), $(p)))\
    echo mkdir_path=$(dir $(word 2, $(pair)))\
    echo cp_path=$(word 1, $(pair)) $(word 2, $(pair));)
$(if ${PRIVATE_PICKUP_FILES}),$(hide) cp -Rf $(PRIVATE_PICKUP_FILES) $(PRIVATE_STAGING_DIR))
# Generate the dict.
$(hide) echo "# For all accepted properties, see BuildImage() in tools/releasetools/build_image.py" > $(PRI
$(hide) echo "# For all accepted properties, see BuildImage() in tools/releasetools/build_image.py" > $(PRI

```

```

inputsource: platform 'amlogic'
[ 97% 219/224] including vendor/sdmc/sdmc-libs/Android.mk ...
hello mtk
vendor/sdmc/sdmc-libs/libmulti_wifi/multi_wifi_load/Android.mk:3: warning: BUILD_BCM_TCP_SH is
[ 98% 225/228] Build image out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/xulantian.img
xulantian_debug_tag
device/amlogic/ohm/xulantian/xulantian1.txt:out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/xulantian /xulantian1.txt
mkdir_path=out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/
cp_path=device/amlogic/ohm/xulantian/xulantian1.txt out/target/product/ohm/obj/PACKAGING/xulantian_intermediates/xulantian
mkdir_path=
cp_path=xulantian1.txt
[ 99% 226/228] Build image out/target/product/ohm/obj/PACKAGING/oem_intermediates/oem.img
xulantian_debug_tag
device/amlogic/ohm/oem/oem.prop:out/target/product/ohm/obj/PACKAGING/oem_intermediates/oem/oem.prop device/amlogic/common/audio/mbox/audio_policy_configuration.xml:out/target
/product/ohm/obj/PACKAGING/oem_intermediates/oem/etc/audio_policy_configuration.xml
mkdir_path=out/target/product/ohm/obj/PACKAGING/oem_intermediates/oem/
cp_path=device/amlogic/ohm/oem/oem.prop out/target/product/ohm/obj/PACKAGING/oem_intermediates/oem/oem.prop
mkdir_path=out/target/product/ohm/obj/PACKAGING/oem_intermediates/oem/etc/
cp_path=device/amlogic/common/audio/mbox/audio_policy_configuration.xml out/target/product/ohm/obj/PACKAGING/oem_intermediates/oem/etc/audio_policy_configuration.xml
[100% 228/228] build out/target/product/ohm/oem.img

### build completed successfully (02:49 (mm:ss)) ###

xulantian@rd24: ~/workspace/s905x4/s/s905x4_sdk8805_reference_userdebug/sdmc/s-amlogic-source-20220805-gtvs.xml$

```

3. 常用外设配置

文件配置：

1.分区表的修改

device/amlogic/\${(product)}/ part_table.txt

//partitionName//startsddr//size//partitionGap//attr

